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# SlideScholar

A Multimodal RAG-Based AI Study Assistant for Lecture Slides

*Project Pre-Proposal · STATGR5293 GenAI Using LLMs · Columbia University · Spring 2026*

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**Abstract —** *Students in lecture-heavy graduate courses spend 3–5 hours weekly converting slides into study materials. Current AI tools handle this poorly: they discard all visual content (diagrams, equations, charts) and are not grounded in the student's actual slides. We propose* ***SlideScholar****, a zero-cost multimodal RAG system that ingests PDF/PPTX lecture slides, understands both text and visuals via LLaVA-7B, indexes per-slide chunks in FAISS, and generates study guides, flashcards, practice exams, and ELI5 explanations grounded exclusively in the student's own slides. A novel adaptive gap detection loop re-retrieves source slides on wrong answers, implementing evidence-based retrieval practice automatically. All computation runs on free-tier GPU infrastructure (Colab Pro + HuggingFace Spaces) at zero cost.*

## 1. CLARITY OF OBJECTIVES

**Research Question:** Can a multimodal RAG system grounded in a student's own lecture slides produce study materials that are more faithful, grounded, and useful than a general-purpose LLM?

### Specific Problems

**Visual Content Lost.** Standard PDF parsers extract raw text only; all diagrams, equations, and charts are silently discarded. STEM lecture slides are overwhelmingly visual, and no existing free tool (ChatPDF, Notion AI) addresses this gap.

**Hallucination Risk.** General-purpose LLMs are not grounded in the student's actual slides. Generated content is unverifiable and frequently contradicts the source material, making it unsuitable for exam preparation.

**No Cross-Lecture Retrieval.** No free tool lets a student query across multiple lecture weeks. Concepts from week 2 cannot be connected to material introduced in week 6, preventing integrated understanding.

### Expected Outcomes & Deliverables

**Working system:** Upload PDF/PPTX → study guide, flashcards, practice exam – all grounded in the student's own slides.

**Live public demo:** Deployed on HuggingFace Spaces (free T4 GPU, public URL).

**Quantitative eval:** RAGAS – faithfulness $\geq$0.80, relevance $\geq$0.75, context recall $\geq$0.70.

**Human study:** 5+ classmates compare SlideScholar vs. ChatGPT baseline on identical slides.

**Ablation study:** Text-only vs. multimodal chunking; slide-level vs. fixed-size segmentation; embedding model comparison.

### Alignment with Course Objectives

- Retrieval-Augmented Generation (RAG) pipelines
- Multimodal LLMs and vision-language models (LLaVA-7B)
- Prompt engineering & structured output formatting
- LLM evaluation frameworks (RAGAS)
- Responsible GenAI: grounding & hallucination reduction

## 2. FEASIBILITY

### Technical Architecture

The system pipeline runs in five sequential stages: (1) **Ingest** – `pdfplumber` and `python-pptx` extract raw text and render each slide as a PNG image; (2) **Understand** – LLaVA-7B generates a natural-language visual description per slide; (3) **Index** – merged text+vision chunks are embedded with `e5-large-v2` and stored in FAISS; (4) **Retrieve** – top-$k$ cosine-similar chunks are fetched at query time; (5) **Generate** – Mistral-7B produces the requested study material, strictly grounded in retrieved context.

| | |
|---|---|
| **Vision** | `LLaVA-7B (4-bit, bitsandbytes)` |
| **Embeddings** | `e5-large-v2 (HuggingFace)` |
| **Vector DB** | `FAISS IndexFlatIP / IVFFlat` |
| **Generation** | `Mistral-7B-Instruct (4-bit)` |
| **UI** | `Gradio (4 tabs: guide, cards, exam, ELI5)` |
| **Eval** | `RAGAS (faithfulness, relevance, recall)` |

### Timeline & Milestones

| | |
|---|---|
| **W1** | Slide parser (PDF + PPTX), per-slide PNG rendering |
| **W2** | LLaVA-7B vision captions, FAISS index build |
| **W3** | RAG retrieval + 4 generation modes (guide, cards, exam, ELI5) |
| **W4** | Gap detection loop, RAGAS eval harness, Gradio UI |
| **W5** | HuggingFace deployment, baseline comparison, final report |

### Resources & Availability

**Colab Pro A100:** Runs LLaVA-7B + Mistral-7B with 4-bit quantization at zero marginal cost.

**HuggingFace Spaces:** Free T4 GPU tier for public demo; all models are open-weight.

**Total budget:** **$0** beyond existing Colab Pro subscription.

### Challenges & Mitigation

**LLaVA-7B GPU memory** → 4-bit quant via `bitsandbytes`; batch slides to cap peak VRAM usage.

**HF Spaces 16 GB RAM cap** → Replace `IndexFlatIP` with FAISS `IVFFlat` (∼60% less memory).

**Model cold-start latency** → Preload all models at app startup; show a Gradio progress bar.

### Scope Definition

**In Scope**
- PDF + PPTX ingestion
- Text + visual chunking
- 4 generation output modes
- Cross-lecture retrieval
- RAGAS evaluation

**Out of Scope (v1)**
- Real-time slide sync
- User accounts / auth
- Non-English slides
- Fine-tuning the LLM

## 3. INNOVATION & RELEVANCE

### Originality of Approach

**Multimodal Slide Understanding.** LLaVA-7B processes every slide *as an image*, generating a natural-language description of all diagrams, equations, and charts. This description is merged with extracted text into one chunk – so retrieval sees the full semantic content of each slide, including visual elements that text-only tools silently discard.

**Slide-Aware Semantic Chunking.** Each slide is one indivisible semantic unit, never split at an arbitrary token boundary. Chunks carry metadata (lecture name, week, slide index) enabling faithful, traceable source citation in every generated output.

**Adaptive Gap Detection Loop.** Wrong practice answers trigger a retrieval step: relevant source slides are fetched and a targeted re-explanation is generated. This *retrieval practice + corrective re-teaching* loop is grounded in cognitive science (spaced retrieval) and is absent from every existing open-source EdTech tool.

### Comparison with Existing Tools

| Feature | ChatPDF | ChatGPT | SlideScholar |
|---|---|---|---|
| Visual understanding | × | ∼ | ✓ |
| Grounded in slides | ✓ | × | ✓ |
| Cross-lecture retrieval | × | × | ✓ |
| Gap detection loop | × | × | ✓ |
| Free / open-source | × | × | ✓ |

### GenAI Research Trends

**Multimodal RAG** Vision-augmented retrieval outperforms text-only RAG on document-understanding benchmarks – active research at Microsoft Research & Google DeepMind (2024).

**Grounded Gen.** Hallucination reduction via faithful retrieval is a central open problem in applied LLM deployment; our RAGAS thresholds operationalise this goal with quantitative targets.

**EdTech AI** LLMs for personalised learning is a high-growth application domain (Khanmigo, Coursera AI, Duolingo Max). This project contributes a free, open-source implementation deployable by any university at $0.

### Potential Impact & Novel Methodology

**Impact.** Any student in any lecture-based course is a direct user. SlideScholar is fully open-source and deployable at zero incremental cost by any university. The adaptive gap detection loop implements evidence-based pedagogy (retrieval practice + targeted re-teaching) in a fully automated pipeline – making high-quality tutoring accessible without human intervention or subscription fees.

**Novel methodology.** The combination of (i) slide-level multimodal chunking, (ii) cross-lecture RAG retrieval, and (iii) an adaptive re-teaching loop triggered by incorrect answers is not present in any existing open-source EdTech system. The full end-to-end pipeline – *ingest* → *understand* → *index* → *retrieve* → *generate* → *re-teach* – is a novel application of GenAI to the EdTech domain, contributing both a working system and a reusable open-source framework.